

FULL APPLICATION AND DEVELOPMENT

#SKILL - 10 → React State Management using useState Object

Prerequisites:

- Basic Idea of JavaScript
- Knowledge of React fundamentals

An online academic portal wants to maintain a list of **students** where the user can **add new students**, **display the list**, and **delete a student** instantly without refreshing the page. The application should use **React Hooks** to manage data at the component level so that the UI updates automatically whenever the state changes.

Your task is to implement a React component using **useState** to store an **object array**, add new items, display them, and delete them dynamically.

1. Create a new React component named **StudentManager**.
2. Inside the component, create an initial **students** array with at least 5 objects containing:
 - a. **id**
 - b. **name**
 - c. **course**
3. Use **useState** to store:
 - a. **students** (array of student objects)
 - b. **newStudent** (object with fields like id, name, course)
4. Create **input boxes** to accept id, name, and course.
 - i. Update the **newStudent** object using **onChange** events.
5. Create an **Add Student** button.
 - i. On click, add the **newStudent** object to the students array.
 - ii. After adding, **clear the input fields**.
6. Display the list of students below using the **students** state.
7. Each student row should have a **Delete** button.
 - i. On clicking delete, remove the record from the students array.
 - ii. The UI should update immediately using **setStudents**.
8. If the list becomes empty, display the message:
"No students available"
9. Add **basic CSS styling** for a clean UI (margin, padding, table style, buttons).
10. Push the React project to **GitHub**.

Create Project in VS code Using Vite

Open terminal and run:

```
npm create vite@latest student-manager
```

Select:

Framework : React
Variant : JavaScript

In bash run:

```
cd student-manager  
npm install  
npm run dev
```

Project Structure

```
student-manager  
├──  
├── src  
│   ├── components  
│   │   ├── StudentManager.jsx  
│   │   └── StudentManager.css  
│   ├── App.jsx  
│   └── main.jsx
```

StudentManager.jsx

```
import React,{useState} from "react";  
import "./StudentManager.css";  
  
function StudentManager(){  
  const initialStudents=[  
    {id:2400031000,name:"Rahul",course:"CSE"},  
    {id:2400031001,name:"Sneha",course:"AI&DS"},  
    {id:2400031002,name:"Arjun",course:"IT"},  
    {id:2400031003,name:"Priya",course:"CS"},  
    {id:2400031004,name:"Kiran",course:"CSE"}  
  ];  
  
  const [students,setStudents]=useState(initialStudents);
```

```

const [newStudent,setNewStudent]=useState({
  id:"",
  name:"",
  course:""
});

const handleChange=(e)=>{
  const {name,value}=e.target;
  setNewStudent({...newStudent,[name]:value});
};

const addStudent=()=>{
  setStudents([...students,newStudent]);
  setNewStudent({id:"",name:"",course:""});
};

const deleteStudent=(id)=>{
  const updated=students.filter((student)=>student.id!==id);
  setStudents(updated);
};

return(

{students.length===0?(

export default StudentManager;

```

StudentManager.css

```

body {

margin: 0;

height: 100vh;

display: flex;

justify-content: center;

align-items: center;

background: linear-gradient(135deg, #eef2f7, #d9e4f5);

font-family: "Segoe UI", Tahoma, Geneva, Verdana, sans-serif;}

.container {

width: 650px;

```

```
background: #ffffff;
padding: 30px;
border-radius: 12px;
box-shadow: 0 10px 25px rgba(0,0,0,0.08);
text-align: center;}

.container h2 {
margin-bottom: 20px;
color: #2c3e50;}

.form {
display: flex;
justify-content: center;
flex-wrap: wrap;
gap: 10px;
margin-bottom: 25px;}

input {
padding: 10px;
width: 150px;
border-radius: 6px;
border: 1px solid #dcdcdc;
font-size: 14px;}

input:focus {
outline: none;
border-color: #4a90e2;}

button {
padding: 10px 16px;
border: none;
border-radius: 6px;
background: #4a90e2;
```

```
color: white;

font-weight: 600;

cursor: pointer;}

button:hover {

background: #357ab8;}

.delete {

background: #e74c3c;}

.delete:hover {

background: #c0392b;}

table {

width: 100%;

border-collapse: collapse;}

th {

background: #4a90e2;

color: white;

padding: 10px;}

td {

padding: 10px;

border-bottom: 1px solid #eee;}

tr:nth-child(even) {

background: #f7f9fc;}

.empty {

margin-top: 15px;

color: #888;}
```

App.jsx

```
import StudentManager from "./components/StudentManager";

function App() {
```

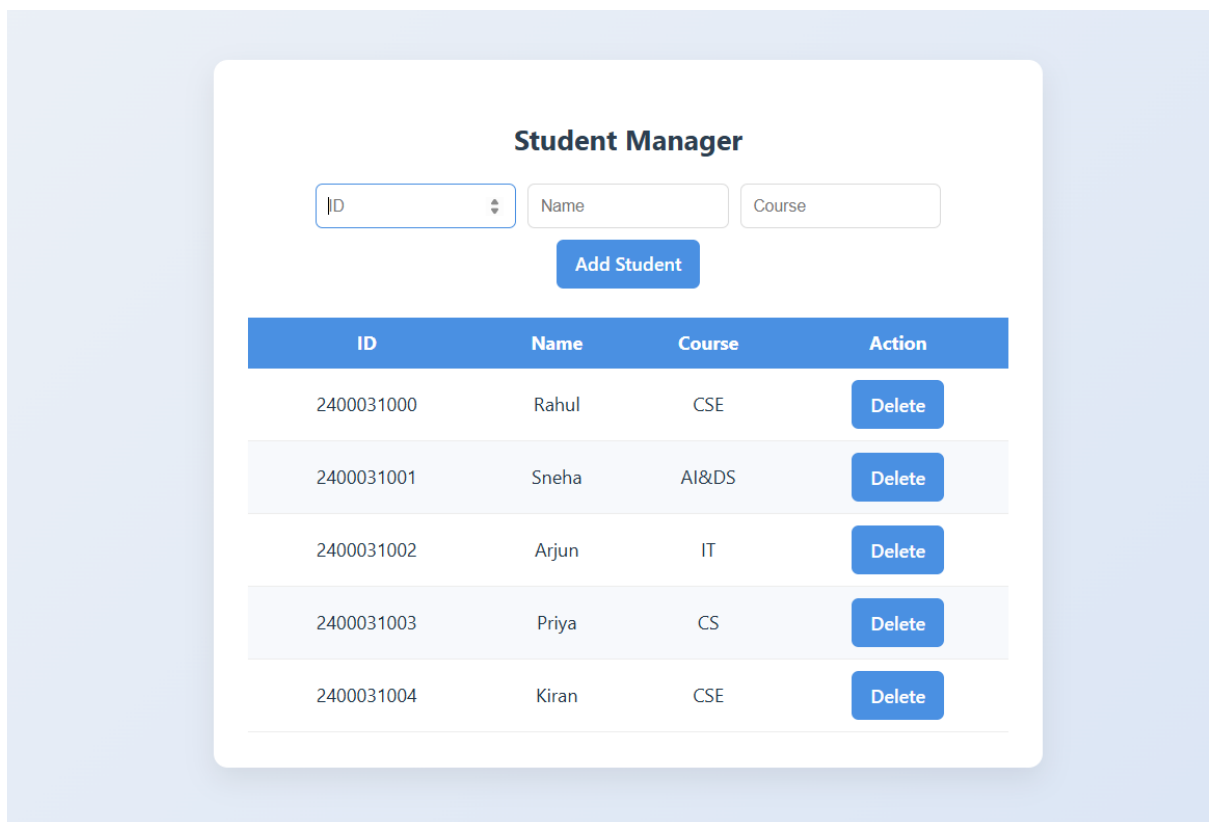
```
return (  
  <>  
  <StudentManager />  
  </> ); }  
export default App;
```

=====

Run the Application

1. Open terminal in project folder: `npm run dev`
 2. Open browser: `http://localhost:5173`
- =====

Output:



The screenshot displays the 'Student Manager' web application. At the top, there is a form with three input fields: 'ID' (containing '2400031000'), 'Name' (containing 'Rahul'), and 'Course' (containing 'CSE'). Below these fields is a blue 'Add Student' button. Underneath the form is a table with four columns: 'ID', 'Name', 'Course', and 'Action'. The table contains five rows of student data, each with a 'Delete' button in the 'Action' column.

ID	Name	Course	Action
2400031000	Rahul	CSE	Delete
2400031001	Sneha	AI&DS	Delete
2400031002	Arjun	IT	Delete
2400031003	Priya	CS	Delete
2400031004	Kiran	CSE	Delete